Shakil Ahmed Rafi

Ph.D. Candidate in Applied Mathematics
Aspiring Data Scientist & M.L. Engineer
191-4 South Hill Ave., Fayetteville, AR, 72701

sarafi@uark.edu
+1 770 686 7234
LinkedIn.com/In/shakilrafi
GitHub.com/2shakilrafi
ShakilRafi.online

Passionate and hard-working graduate student working at the intersection of machine learning, deep learning, and differential equations. I aim to use the tools of mathematics to bring value and make a difference in my community.

Education

University of Arkansas, Ph.D.
Applied Mathematics. *2020-Present.*
Recipient, Lawrence Jesser Toll Jr. Endowed Scholarship.
Treasurer, Graduate Student Colloquium.
Fmr. Vice President, Bangladesh Student Organization.

University of Arkansas, M.Sc.
Pure Mathematics. *2016-2019.*
Recipient, Bangladesh-Sweden Travel Fund.

Troy University, B.Sc.
Pure Mathematics & Philosophy. *2011-2015.*
Graduated Cum Laude
Member, Alabama Eta Chapter of Pi Mu Epsilon, the National Math Honor Society.
Recipient, Millenium Scholar's Award.

Technical Skills

Experienced in Python, including: Scikit-Learn, XGBoost, Pandas, Folium, GeoPandas, MatPlotLib, Keras, NumPy, PyTorch, , and other tools.

Proficient in SQL, Microsoft Azure, PowerBI, and Git.

Certified Wolfram Technology Associate.
Skilled in MATLAB.

Proficient in Pyomo, including solvers such as CBC, Ipopt, bonmin, AMPL, and others.

Proficient in R, R-Studio, tidyverse, Posit Workbench, and Shiny.

Employment History

Intern, Health Economics, Arkansas Blue Cross & Blue Shield
Anticipated, Summer 2023
Used Python to model birth and diabetes outcomes as a Markov Model. Trained neural nets to predict birth outcomes from pre-existing conditions.

Teaching Assistant, Department of Data Science Sam M. Walton College of Business, University of Arkansas *2023—Present*
Assisting a class on Optimization Methods, and a course on Introduction to Data Science Methods.

Graduate Teaching Assistant & Ph.D. Candidate, University of Arkansas
2016-2019,.2019-Present.
Responsible for teaching, lesson-planning, and interfacing with a cohort of faculty in teaching 500+ students along with grading and office hours.

Asst. Business Analyst, R.S.S. Wears Ltd.
2015-2016.
Implemented software for a garment manufacturing start-up, including, payroll, book- keeping with Excel, website setup, paper-trail, and staff directories.

Conferences & Workshops

SIAM Conference on Uncertainty Quantification, Trieste, Italy
Presenting at the upcoming UQ24 conference on how neural networks may accurately approximate functions with reasonable parameter counts.

OAK Supercomputing Fall 2023, Little Rock, AR
SOM-where in Chicago, Rafi
Presented research on Chicago ride-shares analyzing them with self-organizing maps (Kohonen maps)

Simons Laufer Mathamatical Sciences Insitute (formerly MSRI) Summer Graduate School in Machine Learning, San Diego, CA
Intensive summer school focusing on the cutting edge of machine learning.

SIAM Mathematics of Data Science Conferences, 2022, San Diego, CA
Who Rides Uber Anyway? — Rafi & Nithila
We present a clustering approach to classifying ride shares in Chicago during 2020 and the insights it gave us. Awaiting publication.

Arkansas Summer Research Institute, 2022, Online
Who and Where did Loans Go During the Pandemic? — Rafi
We looked at loans disbursed during the pandemic (PPP) and saw that poorer neighborhoods tended to get smaller and less frequent loans.
Funded by Arkansas NSF EPSCOR and DART.

Employment History — contd.

Concessions Manager, Sodexo Inc.
2011-2013
Set up concessions for Troy University games, with a team of six other people, including logistics, safety and customer satisfaction.

Certifications

NVIDIA Deep Learning Institute Fundamentals of Deep Learning
NVIDIAInc.
A certification on the use of NVIDIA tools for the most recent deep learning methodologies.

IBM Data Science Professional
IBM with Coursera, Inc.
A Coursera based course issued by IBM in collaboration with Coursera showing the fundamentals of data science as well as culminating in a real-life project predicting the efficacy of SpaceX launches.

Advanced Google Analytics
Google, Inc.
Advanced tools and techniques of the Google Analytics platform.

Certified Wolfram Technology Associate
Wolfram Technologies, Inc.
Fundamentals of Wolfram Mathematica and the Wolfram Cloud Platform.

Talks Given

Topology & Computation, 2020
Graduate Student Colloquium A survey about current results in the field of computational complexity results as it relates to problems in topology.

Heegard Splittings and Basic Theorems about them, 2020
Graduate Student Colloquium Expository talk about Heegard Splitting, classical results such as Waldhausen's. Heegard Genus, and modern results by Schultens, Rieck et. al.

Why Knot Theorists are Unsure about Tying Shoelaces *Graduate Student Colloquium* Expository talk about Heegard Splitting, classical results such as Waldhausen's. Heegard Genus, and modern results by Schultens, Rieck et. al.

Projects Workshops & Talks —contd.

Arkansas Spring Lecture Series, 2022, Online
Analysis of Ride-Sharing Data for Chicago During the Era of the Pandemic — Rafi & Nithila
We looked at loans disbursed during the pandemic (PPP) and saw that poorer neighborhoods tended to get smaller and less frequent loans.
Funded by NSF and the University of Arkansas.

Service

Vice President of the Bangladesh Student Organization
An organization dedicated to disseminating Bangladeshi culture and helping students of Bangladeshi origin on campus to thrive and feel at home.

Departmental Representative, Graduate Student Council
Representing the interests of the students of the Department of Mathematical Sciences and the Fulbright College of Arts and Sciences.

Treasurer, Graduate Student Colloquium
Responsible for securing funding for speakers when they may appear at our weekly and bi-weekly talks at the Department of Mathematical Sciences.

Conferences Attended

Redbud Topology Conference, 2019, 2020
Representing three states, Arkansas, Missouri, and Oklahoma dedicated to disseminating new and exciting advances in topology, particularly geometric and low-dimensional topology.

Spring Sectional Meeting of the American Mathematical Society, 2018
Representing the interests of the students of mathematics in the states of Arkansas and Oklahoma of the Southern Section of the American Mathematical Society.